

The Reading Room

A book that is never sent whole

A sample book of bdf's secure-reader example

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1. Opening the book

You logged in, and the server knows which books are yours. It does not send this one as a file: it sends ten pages at a time, as you come to them.

Each request carries a public key that your browser made for that request alone. The private key never leaves the page's worker, and WebCrypto will not hand it out.

This is page 3. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

Opening the book, continued

Each request carries a public key that your browser made for that request alone. The private key never leaves the page's worker, and WebCrypto will not hand it out.

This is page 4. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

Opening the book, continued

You logged in, and the server knows which books are yours. It does not send this one as a file: it sends ten pages at a time, as you come to them.

This is page 5. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

Opening the book, continued

Each request carries a public key that your browser made for that request alone. The private key never leaves the page's worker, and WebCrypto will not hand it out.

This is page 6. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

Opening the book, continued

You logged in, and the server knows which books are yours. It does not send this one as a file: it sends ten pages at a time, as you come to them.

This is page 7. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

2. What a segment is

A segment is a small bdf document: every page of the book is in it, with its size, but only ten of them have anything to draw. Its parts are those ten pages need.

Fonts and pictures that several pages share come with the first segment that needs them. The next request says which segments you hold, and the server leaves them out.

This is page 8. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

What a segment is, continued

Fonts and pictures that several pages share come with the first segment that needs them. The next request says which segments you hold, and the server leaves them out.

This is page 9. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

What a segment is, continued

A segment is a small bdf document: every page of the book is in it, with its size, but only ten of them have anything to draw. Its parts are those ten pages need.

This is page 10. It travels in segment 1, with pages 1 to 10, sealed for the request that asked for it.

What a segment is, continued

Fonts and pictures that several pages share come with the first segment that needs them. The next request says which segments you hold, and the server leaves them out.

This is page 11. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

What a segment is, continued

A segment is a small bdf document: every page of the book is in it, with its size, but only ten of them have anything to draw. Its parts are those ten pages need.

This is page 12. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

3. Sealed for one request

The server makes a key pair of its own for each segment, agrees a key with yours, wraps the segment's content key with it, and forgets its private key before it answers.

Your worker agrees the same key, opens the segment, and drops its private key too. After that no one can open what was sent, not even the server.

This is page 13. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

Sealed for one request, continued

Your worker agrees the same key, opens the segment, and drops its private key too. After that no one can open what was sent, not even the server.

This is page 14. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

Sealed for one request, continued

The server makes a key pair of its own for each segment, agrees a key with yours, wraps the segment's content key with it, and forgets its private key before it answers.

This is page 15. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

Sealed for one request, continued

Your worker agrees the same key, opens the segment, and drops its private key too. After that no one can open what was sent, not even the server.

This is page 16. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

Sealed for one request, continued

The server makes a key pair of its own for each segment, agrees a key with yours, wraps the segment's content key with it, and forgets its private key before it answers.

This is page 17. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

4. What a recording holds

Someone who recorded the traffic holds sealed segments. Stealing the server's TLS key later, or your password, or your session cookie, opens none of them.

The session only lets its holder ask for new segments, as you could: the server checks your rights, and how fast pages are read, before it seals anything.

This is page 18. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

What a recording holds, continued

The session only lets its holder ask for new segments, as you could: the server checks your rights, and how fast pages are read, before it seals anything.

This is page 19. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

What a recording holds, continued

Someone who recorded the traffic holds sealed segments. Stealing the server's TLS key later, or your password, or your session cookie, opens none of them.

This is page 20. It travels in segment 2, with pages 11 to 20, sealed for the request that asked for it.

What a recording holds, continued

The session only lets its holder ask for new segments, as you could: the server checks your rights, and how fast pages are read, before it seals anything.

This is page 21. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What a recording holds, continued

Someone who recorded the traffic holds sealed segments. Stealing the server's TLS key later, or your password, or your session cookie, opens none of them.

This is page 22. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

5. What stays on the device

Nothing is written to the disk: the answers are sent with Cache-Control no-store, and the keys live in the worker's memory while the book is open.

Close the tab, or log out, and there is nothing left to open. This site does not read books offline; that would need a key kept on the device.

This is page 23. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What stays on the device, continued

Close the tab, or log out, and there is nothing left to open. This site does not read books offline; that would need a key kept on the device.

This is page 24. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What stays on the device, continued

Nothing is written to the disk: the answers are sent with Cache-Control no-store, and the keys live in the worker's memory while the book is open.

This is page 25. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What stays on the device, continued

Close the tab, or log out, and there is nothing left to open. This site does not read books offline; that would need a key kept on the device.

This is page 26. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What stays on the device, continued

Nothing is written to the disk: the answers are sent with Cache-Control no-store, and the keys live in the worker's memory while the book is open.

This is page 27. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

6. What this does not do

A reader who may read the book can still copy what the screen shows. What the server can do is decide who reads which pages, and how fast.

The server keeps the book in the clear, converted once. Whoever takes the server's disk takes the books; that is a matter for the server, not for the wire.

This is page 28. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What this does not do, continued

The server keeps the book in the clear, converted once. Whoever takes the server's disk takes the books; that is a matter for the server, not for the wire.

This is page 29. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What this does not do, continued

A reader who may read the book can still copy what the screen shows.

What the server can do is decide who reads which pages, and how fast.

This is page 30. It travels in segment 3, with pages 21 to 30, sealed for the request that asked for it.

What this does not do, continued

The server keeps the book in the clear, converted once. Whoever takes the server's disk takes the books; that is a matter for the server, not for the wire.

This is page 31. It travels in segment 4, with pages 31 to 40, sealed for the request that asked for it.

What this does not do, continued

A reader who may read the book can still copy what the screen shows.

What the server can do is decide who reads which pages, and how fast.

This is page 32. It travels in segment 4, with pages 31 to 40, sealed for the request that asked for it.